

Programming Assignment # 3, Eigen Faces

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Due Date: 05 Nov 2025

1 Objective

- In this assignment you will learn to apply Principal Component Analysis (PCA) for the data reduction. You will download images of faces of known actors dataset and apply PCA for the reducing the size of the data. Then you will apply some classification method for detection of images.

2 Dataset

For this assignment you will use the cropped version of the Labeled Faces in the Wild (LFW) dataset that has kept the faces of subjects only, while omitting the background. This dataset can be downloaded from <https://conradsanderson.id.au/lfwcrop/>. You can also download/load the data from Kaggle. It has the gray scale and colored images. You only have to download the gray scale images. The split of the dataset is given description file.

3 Application of Principal Component Analysis

One of the applications of PCA is in image compression. Assuming that the matrix X is of size $m \times n$ i.e., an image, $\mathbb{R}^{m \times n}$. The principal components (U) returned by any optimization technique will be of size $m \times m$ i.e., $\mathbb{R}^{m \times m}$. Recall also that the columns of U are the principal component directions, and sorted in decreasing order of ‘importance’ or ‘principality’, so that the ‘most principal’ component is the first column of U .

Suppose that we truncate the matrix U before projecting the data using the $Z = U^T \times X$. This new matrix U_{reduce} is obtained by $U_{reduce} = U(:, 1 : k)$ where $k < m$ and $U_{reduce} \in \mathbb{R}^{m \times k}$. Now the new projected data obtained by $\hat{Z} = U_{reduce} \times X$ which is still dimensionally correct and consistent producing the matrix $\hat{Z} \in \mathbb{R}^{m \times k}$.

Now let us try to regenerate the data by using $\hat{X} = \hat{U} \times \hat{Z}$. The newly recovered data has the original dimensions of matrix X , $\hat{X} \in \mathbb{R}^{m \times m}$. Although, the matrix X and $\hat{X}$ are of same dimensions but they are not exactly same as we had reduced the size of U -matrix before data projection. So we have actually say that the reproduced matrix $\hat{X}$ has lost some information.

Things to Do

1.
 - Take an interesting dataset (explore Kaggle as well)
 - Generate Python code that uses PCA for reducing the dimensionality of the data set

- Plot points (3 Principal Components) as a 3-D Graph
- You can use the following datasets: IRIS Dataset, Wine Dataset, etc.

Note: You can use ChatGPT to generate this code.

2.
 - Download the LFW dataset from the given website.
 - Implement PCA for reducing the size of images.
 - The data is already split into train and test, see the README.txt and folder titled 'lists'.
 - Using any classifier of your choice, get the classification accuracy on the images.
 - Apply PCA on the data and keep 97% of variance captured by the principal components. Find the classification accuracy.
 - Apply PCA on the data and keep 95% of variance captured by the principal components. Find the classification accuracy.
 - Plot a graph of variance kept vs the accuracy.
 - Plot a graph of variance kept vs the # components kept.

4 What to Submit

Report: You need to submit a report that contains all the graphs and your methodology that you have taken to implement and obtain each graph. You can support your arguments by adding snippets of your code (not the whole code).

Note: This is an easy problem so it should not take you very long to complete.

WARNING:

- Try not to copy code or report. Anybody caught doing so, will get zero marks.